Technical Overview: Secure Online Voting System

Prepared by: Minenhle Njabulo Trinity Dlamini

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1. Introduction

This document presents a technical overview of a proposed Online Voting System designed to support national, provincial, and municipal elections in South Africa. The system prioritises security, transparency, auditability, and accessibility, while aligning with requirements for public trust and large-scale election management.

2. System Architecture

The system is built using a modular, service-oriented architecture consisting of:

- Authentication & Identity Service
- Voting Service
- Audit & Logging Service
- Results Publication Service
- Admin & Oversight Dashboard
- Voter Mobile App (React Native)
- Voter Web App (React)

3. Security Framework

Voter Authentication includes multi-factor authentication using ID + phone verification, OTP, and optional biometrics. Votes are encrypted on-device using RSA-4096 and AES-256-GCM. Anti-fraud measures include hash fingerprints, tamper detection, and anomaly detection.

4. Voting Process Workflow

1. Voter logs in.
2. Registration verified.
3. Ballot issued.
4. Voter selects choices.
5. Vote encrypted locally.
6. Encrypted vote submitted.

7. Hash + timestamp logged.

8. Confirmation returned.

5. Real-Time Public Results Engine

Results refresh every 10 minutes. Only aggregated counts are shown. A read-only replica server stores tallies.

6. Transparency & Auditing

Audit trails include timestamps, hashes, device signatures, and logs. Public audit layer supported via blockchain or hash-chain logs.

7. Accessibility & User Experience

Web portal (React), mobile app (React Native), WCAG compliant, low-data optimized.

8. Scalability & Performance

Supports 20–30 million voters. Horizontal scaling via Kubernetes/Docker. Deployable on IEC infrastructure, government cloud, or hybrid.

9. Implementation Roadmap

Phase 1 – Prototype (Complete)

Phase 2 – IEC Review & Security Audit

Phase 3 – Pilot Program

Phase 4 – National Rollout

10. Conclusion

This system is secure, transparent, and built for South Africa's future elections. A live demonstration is available upon request.